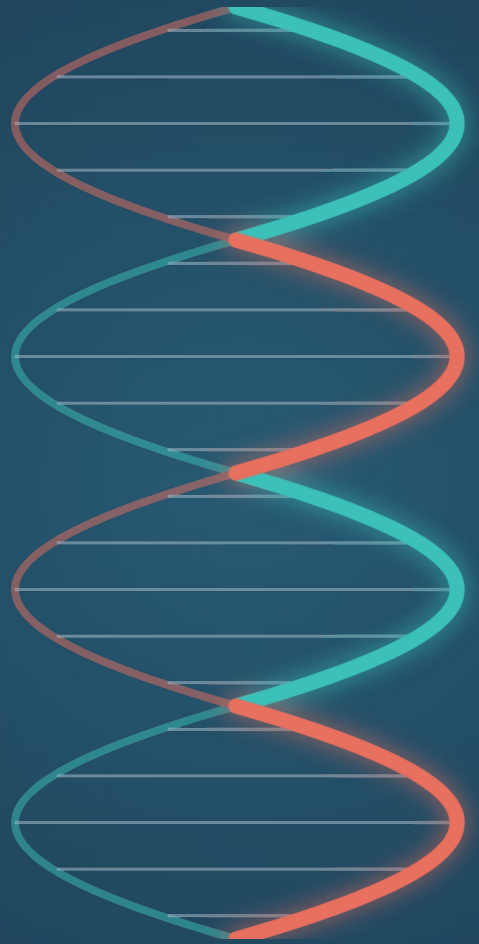
S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

Helix

From financial text to measurable risk

Financial text → company signals → portfolio + bank risk



0.821
Company sentiment F1
Conflicting SEntFiN rows

0.661
Credit-gate precision
E23 design data

0.669
Event macro-F1
E44b AI-agreed news

Anubhav Jha · VIT Bhopal

Connect financial headlines to risk decisions

A risk desk needs the company, the event and the financial consequence.

Who is affected?

Separate the company in the news from a bank giving an analyst rating.

What happened?

Distinguish a credit event from earnings, products or market news.

What does it change?

Translate signals into portfolio weights and a bank stress scenario.

STORED REPLAY OUTPUT · 21 APR 2020

“Coca Cola Q1 EPS $0.51 Beats $0.44 Est; Revenue $8.6B Beats $8.28B Est”

Company	Sentiment	Event	Move score
KO	**+0.90**	**Earnings**	**10 / 10**

A · Portfolio rebalancing

Sentiment + prices
→ capped portfolio weights

B · Bank stress testing

Credit / market events
→ loss and capital estimates

One text pipeline. Two risk workflows. Financial calculations stay explicit.

Text goes in; two risk workflows come out

Public inputs

News + social

GDELT live; FNSPID history; stock tweets.

Financial context

Prices, SEC debt, ratings and FRED. Assumed bank book.

Shared NLP

Company linking

15 replay / 99 live names

Two text tasks

Company sentiment + 12-class event output

Impact · 1–10

Batch: learned + prices
Live: formula

Signals and gates

Company signal

Ticker, role, time
Sentiment, event, impact

Credit gate

Obligor + credit evidence

Market gate

Geo / macro events

Risk modules

A · Portfolio

Sentiment tilt; 15% cap. 5 bp cost; equal-weight comparison.

B · Bank stress

Loans, bonds, derivatives
Losses, IFRS 9 and IRB
CET1; reverse stress

Financial inputs feed impact / risk calculations directly

React dashboard
Replay · live feed · analyze

FastAPI interface
REST + replay SSE

Saved signals
CSV / JSON / JSONL

Built to replay, explain and recover

01 One local service

FastAPI serves the React app.
INT8 ONNX models run on CPU.

Docker · browser → /api

02 Live and replay

Live headlines are scored now.
Replay reads saved signals.

Live impact uses the formula

03 Evidence gates

Credit and market gates route signals into bank stress tests.

Exposure inputs stay separate

04 Saved state

CSV/JSONL signals and caches.
Duplicate checks use URL + text.

Recent live context is bounded

05 Recovery and health

/api/live/health tracks feed errors.
Opt-in previews recover on restart.

Live previews: off by default

06 Auditable records

Stored alerts and weights carry SHA-256 links and provenance.

Verify against a trusted copy

Single host · separate live and historical records · scenario previews do not execute trades

Results against explicit baselines

0.821

Company sentiment

macro-F1 · vs 0.476

463 entity rows (conflicts)
SEntFiN · human labels

0.661

Credit gate

precision · vs 0.470

243 pairs · recall 0.848
E23 · AI-labelled design data

0.353

Impact ranking

Spearman ρ · vs 0.331

Fresh stocks, 2020
Development check

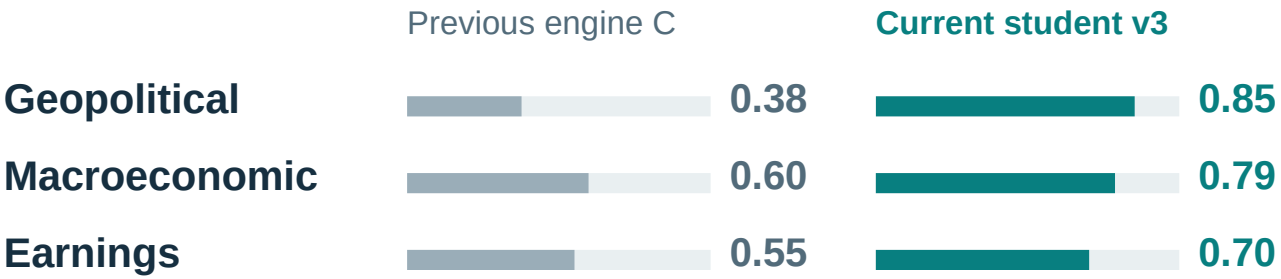
0.669

Event model · v3

macro-F1 · vs 0.604

822 AI-agreed headlines
12 classes · E44b

Largest class gains · F1



E44b · 95% intervals

F1 0.669 [0.635, 0.695]
Gain +0.065 [0.039, 0.088]
78 disagreements excluded; AI labels.
Covenant F1 0.386; Default n=28.
This set also selected v3.

Portfolio · tilt 24.85% vs equal weight 25.52% · placebo p = 0.403 · no proven edge (Apr–Jun 2020).

One downgrade, traced to bank capital

Boeing · 10 Apr 2020 · inspect the credit evidence and capital shortfall

Headline

Moody’s cuts Boeing Baa1 → Baa2

Tweet · 10 Apr 2020

Engine reads

Credit downgrade

Formula severity: 8 / 10

Credit gate

Fired

Rating-agency evidence

Stress scenario

Assumed book notch A− → BBB+

COVID market episode

Scenario outputs · hypothetical bank book

CET1 capital ratio

13.0% → 6.8%

Common Equity Tier 1

Loss before tax

$203.0m

Loan, bond and hedge revaluation

Gap below the 8% scenario floor

1.2 pp

pp = percentage points

Use case · flag the capital shortfall and inspect the obligor exposure.

Book dated 31 Dec 2019 · Boeing downgrade + COVID scenario · assumptions, not a forecast.

Limits and the next validation steps

Limits today

- Events: 0.669 on 822 AI-agreed rows; the same set selected v3.
- Credit: precision is from seen design data, not an unseen alert test.
- Portfolio: no demonstrated edge over equal weight (placebo p ≈ 0.4).
- Bank stress: hypothetical exposures; loss magnitudes are not calibrated.

Next validation step

- A fresh event test, with a human review of the reference labels.
- Unseen credit alerts, including counterparty false positives.
- Longer portfolio tests across market regimes and realistic costs.
- Validate exposure inputs and calibrate scenario losses.

Built with Claude + ChatGPT. New project labels are AI-made, with no human check; SEntFiN supplies human labels.
Public and synthetic data. Risk analytics for demonstration, not investment advice.